

# CORRELATIONS BETWEEN PRIMITIVE ROOT STATUSES OF CONSECUTIVE PRIMES

**ABSTRACT.** Call a prime  $p$  an *Artin prime for base 10* if 10 is a primitive root modulo  $p$ . We investigate the dependence between the Artin statuses of *consecutive* primes. Computing Artin status for all 50,847,531 primes  $7 \leq p \leq 10^9$ , we find that consecutive primes are measurably *anticorrelated*:  $P(\text{Art}_{n+1} \mid \text{Art}_n) = 0.36514$  versus  $P(\text{Art}_{n+1} \mid \neg \text{Art}_n) = 0.37928$ , a deficit  $\delta = -0.01414$  (Pearson statistic 10,167 on one degree of freedom; signed square root  $-100.8$  under a nominal independent-status reference model; over 50,847,530 consecutive pairs). The dependence has striking gap structure. We prove an elementary exclusion law: if  $p, p+g > 5$  are both prime with  $g \equiv 20 \pmod{40}$ , then 10 is a quadratic residue modulo exactly one of them, so they can never both be Artin primes for base 10; empirically, among 2,195,882 consecutive pairs with gap 20 or 60 there is indeed not a single doubly-Artin pair; an independent recomputation extends this to 194,296,748 such pairs below  $10^{11}$ , again with none. Conversely, gaps  $g \equiv 0 \pmod{40}$  force equal quadratic-residue status and exhibit strong positive dependence ( $P(\text{Art}_{n+1} \mid \text{Art}_n) = 0.899$  at  $g = 40$ ). Conditioning on the joint residues of  $(p_n, p_{n+1})$  modulo 120 or 840 substantially reduces a selected-cell signed residual association; the remaining association is small in that metric but is not established to vanish. These results are consistent with a substantial role for arithmetic residue structure in the observed anticorrelation; they do not constitute a complete decomposition or an independent quantitative verification of the Hardy–Littlewood explanation. As an intermediate link we report the (apparently unreported) anticorrelation  $r(\omega(p_n - 1), \omega(p_{n+1} - 1)) = -0.041$ . At four decade-spaced cutoffs  $x = 10^8, 10^9, 10^{10}, 10^{11}$  the deficit grows in magnitude with decreasing increments,  $\delta = -0.01336, -0.01414, -0.01472, -0.01500$ , and no turnover is observed; we draw no asymptotic conclusion from this, and in particular these finite-range values do not distinguish eventual decay to zero from persistence of a nonzero correlation. We formulate open questions and discuss extensions to general bases.

## 1. INTRODUCTION

Two classical themes in the theory of primes intersect in this paper.

The first is Artin’s primitive root conjecture, communicated to Hasse in 1927 (see [1] and the historical account in [15]): for any integer  $a \notin \{-1, 0, 1\}$  that is not a perfect square, the set of primes  $p$  for which  $a$  is a primitive root modulo  $p$  has positive density relative to the primes; for base  $a = 10$  this density is Artin’s constant  $C = \prod_q (1 - \frac{1}{q(q-1)}) \approx 0.3739558$ . Hooley [11] proved the conjecture under the Generalized Riemann Hypothesis. Unconditionally, Gupta and Murty [8] and Heath-Brown [10] showed that the conjecture can fail for at most two prime values of  $a$ , though without an asymptotic for the counting function; in the averaged direction Klurman, Shparlinski and Teräväinen [12] recently established the predicted asymptotic for almost all  $1 \leq a \leq \exp((\log \log x)^2)$ , substantially improving the range of Stephens (1969). Moree’s survey [15] gives a comprehensive account. For  $a = 10$  the property has a pleasant classical meaning: 10 is a primitive root mod  $p$  precisely when the decimal expansion of  $1/p$  has maximal period  $p - 1$  (the *full reptend* or *long* primes; see [3, pp. 166–171]).

The second theme is the surprising dependence between *consecutive* primes quantitatively investigated and conjecturally explained by Lemke Oliver and Soundararajan [14] (earlier numerical observations of such biases are due to Knapowski–Turán, Ko, and Ash–Beltis–Gross–Sinnott, as

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[14] itself discusses): for a modulus  $q$ , a prime in a given reduced residue class mod  $q$  is followed by a prime in the *same* class markedly less often than equidistribution would predict. The phenomenon, which they explain conjecturally via the Hardy–Littlewood prime  $k$ -tuple conjectures [9], decays slowly and is prominent at the scales accessible to computation.

This paper asks a correlational question: *are the Artin statuses of consecutive primes correlated?* The adjacent question of when two primes at a fixed even shift can *both* be Artin primes was taken up by Tinková, Waxman and Zindulka [13], whose Theorem 1.1 identifies shift–base pairs admitting no such pair; the exclusion law of Theorem 1 below is the base-10 instance of that phenomenon and we claim no priority for it. The correlation measured here, over consecutive primes rather than a fixed shift, appears not to have been studied. Since the Artin property of  $p$  is governed by the arithmetic of  $p-1$  (which primes  $q$  divide it), and since the residues of consecutive primes modulo small  $q$  are correlated by [14], one expects *some* induced dependence between  $\mathbf{1}[10 \text{ is a primitive root mod } p_n]$  and  $\mathbf{1}[10 \text{ is a primitive root mod } p_{n+1}]$ . What is not obvious is its sign, its size, its structure as a function of the gap  $g = p_{n+1} - p_n$ , or the extent to which it is connected to known arithmetic structure. We answer these questions computationally for all primes up to  $10^9$  and isolate one component that is not merely statistical but *deterministic*: an exclusion law for gaps  $g \equiv 20 \pmod{40}$  (Theorem 1).

Related but distinct questions have been studied. Garcia, Kahoro, and Luca [6] (see also the sequel [7]) compared the *number* of primitive roots,  $\varphi(p-1)$  versus  $\varphi(p+1)$ , for twin prime pairs; under the Bateman–Horn conjecture they establish a majority bias together with a positive exceptional proportion, and conjecture a precise value for that proportion. Pollack [17] proved (under GRH) that for every nonsquare  $g \neq -1$  and every  $m$ , there are infinitely many runs of  $m$  consecutive primes each having  $g$  as a primitive root. Baker and Pollack [2] proved unconditionally that if  $Q$  is a set of at least  $\exp(Cm)$  multiplicatively independent primes (for an absolute constant  $C$ ), then infinitely many runs of  $m$  consecutive primes each have some element of  $Q$  as a primitive root, with the primes lying in a bounded-length interval. We emphasise that these are *existence* results, produced by sieve constructions that select rare favourable configurations; they make no statement about the typical joint behaviour of Artin status along the sequence of primes. The present paper asks the complementary *statistical* question — how the Artin statuses of consecutive primes are correlated on average — which appears to be unexamined; and the exclusion law of Theorem 1, though elementary, seems to be unrecorded.

Recent work has also sharpened the analytic understanding of Artin densities for *individual* primes: Fan and Pollack [5] establish (under GRH) a version of the Artin–Hooley asymptotic that is uniform in the base, and Perucca and Shparlinski [16] prove (under GRH) uniform upper and lower bounds for the Artin density relative to the corresponding Artin constant. These results concern the distribution of Artin primes one prime at a time; to our knowledge the *joint* behaviour of the Artin property at consecutive primes, which is the subject of this paper, has not been considered.

### Summary of results.

- (1) (Theorem 1) If  $p$  and  $p+g$  are primes with  $p, p+g > 5$  and  $g \equiv 20 \pmod{40}$ , then  $\left(\frac{10}{p+g}\right) = -\left(\frac{10}{p}\right)$ ; consequently 10 is a quadratic residue for exactly one of the two, and at most one can be an Artin prime for base 10. If instead  $g \equiv 0 \pmod{40}$ , then  $\left(\frac{10}{p+g}\right) = \left(\frac{10}{p}\right)$ .
- (2) (Section 4) Across all 50,847,530 consecutive prime pairs with  $7 \leq p_n < p_{n+1} \leq 10^9$ , Artin statuses are anticorrelated:  $\delta = P(\text{Art}_{n+1} \mid \text{Art}_n) - P(\text{Art}_{n+1} \mid \neg \text{Art}_n) = -0.01414$ , with Pearson  $\chi^2 = 10,167$  on one degree of freedom.
- (3) (Section 5) The dependence is violently non-uniform in the gap:  $\delta(g)$  ranges from  $-0.592$  (at  $g = 60$ , where the theorem forces exclusion) to  $+0.643$  (at  $g = 40$ ). The exclusion classes  $g \equiv 20 \pmod{40}$  are forced by Theorem 1; for the remaining classes we observe a

qualitative association between the sign of  $\delta(g)$  and the residues  $g \bmod 40$  and  $g \bmod 3$ , but we do not exhibit a predictor of its magnitude.

- (4) (Section 6) Conditioning on the pair of residues of  $(p_n, p_{n+1})$  modulo 120 or 840 substantially reduces a selected-cell signed residual metric; the remaining small residual is consistent with further small-modulus channels but has not been shown to vanish. See Section 6 for exact statistics and their limitations.
- (5) (Section 7) As an intermediate link we measure  $r(\omega(p_n - 1), \omega(p_{n+1} - 1)) = -0.0410$ : consecutive primes'  $p - 1$  factorisations repel.

Throughout, *Artin prime* means Artin prime for base 10 unless stated otherwise, and  $p_n$  denotes the  $n$ -th prime.

## 2. THE EXCLUSION LAW

**Definition 1.** For an odd prime  $p \nmid 10$ , write  $\left(\frac{10}{p}\right)$  for the Legendre symbol. We say  $p$  is an *Artin prime* (for base 10) if the multiplicative order of 10 modulo  $p$  equals  $p - 1$ .

If 10 is a quadratic residue mod  $p$  then its order divides  $(p - 1)/2$ , so a QR is never a primitive root;  $\left(\frac{10}{p}\right) = -1$  is a necessary (not sufficient) condition for the Artin property.

**Theorem 1** (Exclusion law for gaps  $\equiv 20 \pmod{40}$ ; cf. [13, Thm. 1.1]). *Let  $p$  and  $p' = p + g$  be primes with  $p, p' > 5$  and  $g > 0$ ,  $g \equiv 20 \pmod{40}$ . Then*

$$\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right).$$

*In particular, 10 is a quadratic residue modulo exactly one of  $p, p'$ , and at most one of them is an Artin prime for base 10.*

*Proof.* By multiplicativity,  $\left(\frac{10}{p'}\right) = \left(\frac{2}{p'}\right) \left(\frac{5}{p'}\right)$ .

Since  $g \equiv 0 \pmod{5}$ , we have  $p' \equiv p \pmod{5}$ . By quadratic reciprocity ( $5 \equiv 1 \pmod{4}$ ),  $\left(\frac{5}{p'}\right) = \left(\frac{p}{5}\right)$  depends only on  $p \bmod 5$ ; hence  $\left(\frac{5}{p'}\right) = \left(\frac{5}{p}\right)$ .

Since  $g \equiv 4 \pmod{8}$ , we have  $p' \equiv p + 4 \pmod{8}$ . The map  $r \mapsto r + 4$  on odd residues mod 8 exchanges  $1 \leftrightarrow 5$  and  $3 \leftrightarrow 7$ . By the second supplementary law,  $\left(\frac{2}{p}\right) = +1$  iff  $p \equiv \pm 1 \pmod{8}$ , and the exchange maps  $\{1, 7\}$  onto  $\{5, 3\}$  and vice versa; hence  $\left(\frac{2}{p'}\right) = -\left(\frac{2}{p}\right)$  for every admissible residue.

Multiplying,  $\left(\frac{10}{p'}\right) = -\left(\frac{10}{p}\right)$ . Whichever of the two primes has  $\left(\frac{10}{\cdot}\right) = +1$  satisfies  $\text{ord}(10) \mid (p - 1)/2 < p - 1$ , so it is not an Artin prime.  $\square$

**Corollary 1** (Coupling law for gaps  $\equiv 0 \pmod{40}$ ). *If  $p, p' = p + g$  are primes with  $p, p' > 5$  and  $g \equiv 0 \pmod{40}$ , then  $\left(\frac{10}{p'}\right) = \left(\frac{10}{p}\right)$ . In particular, if either is an Artin prime, then 10 is a quadratic non-residue for both.*

*Proof.* Now  $g \equiv 0 \pmod{8}$  and  $g \equiv 0 \pmod{5}$ , so both  $\left(\frac{2}{\cdot}\right)$  and  $\left(\frac{5}{\cdot}\right)$  are preserved.  $\square$

**Remark 1.** *The modulus 40 is specific to base 10: it is the conductor of the quadratic character  $p \mapsto \left(\frac{10}{p}\right)$ , namely  $4 \cdot 10 = 40$  (the discriminant of  $\mathbb{Q}(\sqrt{10})$ ). For a general non-square integer base  $a$  the same argument applies with 40 replaced by the conductor  $|d_a|$  of the quadratic character attached to  $\mathbb{Q}(\sqrt{a})$ : for odd primes  $p, p'$  with  $p \nmid a$  and  $p' \nmid a$ , gaps  $g \equiv 0 \pmod{|d_a|}$  preserve quadratic-residue status. The coprimality hypothesis is needed, not cosmetic: the primitive character of the field agrees with  $\left(\frac{a}{\cdot}\right)$  only away from the primes dividing  $a$ , and a prime dividing a square factor of*

$a$  gives  $\left(\frac{a}{p}\right) = 0$  while the field character is nonzero. For instance  $a = 98 = 2 \cdot 7^2$  has conductor 8, and the primes 7 and 23 differ by  $16 \equiv 0 \pmod{8}$ , yet  $\left(\frac{98}{7}\right) = 0$  while  $\left(\frac{98}{23}\right) = \left(\frac{2}{23}\right) = +1$ . (The base-10 theorem is unaffected: its hypotheses exclude  $p = 2, 5$ .) Some bases also admit residue classes of gaps that reverse the Legendre symbol on every admissible pair; their existence requires a separate classification. Base 10 is distinguished only by our decimal habit and the full-reptend interpretation.

**Remark 2.** For gaps in classes other than  $0, 20 \pmod{40}$  the relation between  $\left(\frac{10}{p}\right)$  and  $\left(\frac{10}{p'}\right)$  depends on  $p$  as well as  $g$  (the shift on  $p \pmod{8}$  or  $p \pmod{5}$  is not a symbol-preserving or symbol-flipping involution), producing the partial correlations catalogued in Section 5.

We verified Theorem 1 computationally as a guard against error: for all 23,956 consecutive prime pairs below  $10^7$  with gap  $\equiv 20 \pmod{40}$ , the Legendre symbols flip in every single case, and a direct order computation on a systematic subsample found no doubly-Artin pair. In the full dataset up to  $10^9$  (Section 5), the 2,195,882 consecutive pairs with  $g \in \{20, 60\}$  contain *no* pair in which both primes are Artin — an empirical count of zero, exactly as the theorem requires.

### 3. DATA AND COMPUTATION

For every prime  $7 \leq p \leq 10^9$  we compute: the gap to the neighbouring primes, the number  $\omega(p-1)$  of distinct prime factors of  $p-1$  (by trial division), the Artin indicator (computed exactly via  $10^{(p-1)/q} \not\equiv 1 \pmod{p}$  for each prime  $q \mid p-1$ , using GMP modular exponentiation), and small residues of  $p$ . A segmented sieve of Eratosthenes generates the primes; the corrected sieve emits all 50,847,531 primes from 7 to 999,999,937 (the largest prime  $\leq 10^9$ ).

**Independent recomputation.** The census has been recomputed from scratch by a second, deliberately disjoint implementation, which shares no code with the sieve above: it uses hand-written 128-bit modular exponentiation in place of GMP, and it factors  $p-1$  by a segmented sieve of factors over the shifted interval rather than by per-prime trial division, so that an error in either implementation is unlikely to be mirrored in the other. Both were first checked against **sympy** at  $10^6$ , where all three routes agree exactly. At  $10^9$  the independent implementation reproduces the published census in every quantity: 50,847,531 primes, 19,016,617 Artin primes, the joint table  $\begin{bmatrix} 19758045 & 12072868 \\ 12072869 & 6943748 \end{bmatrix}$ ,  $\delta = -0.014140158840$ , and zero doubly-Artin pairs among all 2,195,882 pairs at gaps 20 and 60.

The same program extends the census to  $10^{10}$  (455,052,508 primes) and  $10^{11}$  (4,118,054,810 primes). In both cases the prime count equals  $\pi(x)$  less the three excluded primes 2, 3, 5, and the Artin density agrees with Artin's constant 0.3739558 to six decimal places (0.373955 and 0.3739543). Theorem 1 continues to hold: there are no doubly-Artin pairs among the 20,700,958 pairs at gaps 20, 60 below  $10^{10}$ , nor among the 194,296,748 such pairs below  $10^{11}$ . The computation is run as a few hundred contiguous shards and stitched, with the pairs straddling shard boundaries added back; re-sharding  $10^{10}$  under a different partition reproduces every quantity identically, including  $\delta$  to fifteen decimal places, so the totals do not depend on the shard layout.

**Endpoint correction.** An earlier version of the sieve contained two endpoint bugs: it emitted a duplicate row for  $p = 7$  and omitted the final prime  $p = 999,999,937$ . These have been corrected. The duplicate was already skipped by the analysis scripts; the missing endpoint adds one pair at gap 8 (both primes non-Artin:  $\text{ord}_{10}(999,999,937) = 333,333,312$ ; independently verified via **sympy**). This correction changes the pair count from 50,847,529 to 50,847,530 and does not materially alter any displayed summary statistic.

As a global check, the number of Artin primes in the dataset is 19,016,617, a proportion 0.373993, matching Artin's constant 0.373956 to four decimal places (for base 10 the squarefree part is  $10 \not\equiv 1 \pmod{4}$ , so no correction factor applies and the conjectural density is exactly  $C$ ; see [11, 15]). Agreement with a conjectural density is a sanity check, not a verification of every indicator.

The consecutive-pair analysis is a single streaming pass over the 50,847,530 ordered pairs  $(p_n, p_{n+1})$ , accumulating  $2 \times 2$  contingency tables of  $(\mathbf{1}[\text{Art}_n], \mathbf{1}[\text{Art}_{n+1}])$ : globally, stratified by gap  $g$ , and stratified by joint residues of  $(p_n, p_{n+1})$  to various moduli. All counts reported below are exact (no sampling), except where noted. Analysis scripts and their outputs are available with the data; see the Data Availability statement.

**Runtime, hardware, and validation.** The computations to  $10^9$  were performed on a single machine with a 2-core Intel Xeon E5-2673 v4 processor at 2.3 GHz and 8 GB of RAM, running Ubuntu Linux, using no parallelism beyond a single core. The independent recomputation and the extensions to  $10^{10}$  and  $10^{11}$  were run on a separate 32-core AMD machine (gcc 15.2.0) as a few hundred contiguous shards; that run is described above and uses a different implementation. The segmented sieve (segment length  $2^{21}$ , so the working set stays within cache) together with the per-prime computation — trial-division factorisation of  $p - 1$  and the Artin test by modular exponentiation — generates the full dataset to  $10^9$ ; peak memory usage is under 200 MB, as only one segment and the small primes to  $\sqrt{x}$  are held in memory. The streaming pair analysis reads the dataset once and runs in minutes. As validation we checked: (i) the prime count  $\pi(10^9) = 50,847,534$  against the known value; (ii) the Artin prime count 19,016,617 (proportion 0.373993) against Artin’s constant 0.373956, as discussed above; (iii) Theorem 1 on all 23,956 pairs with gap  $\equiv 20 \pmod{40}$  below  $10^7$ , with an independent `sympy` order computation on a subsample; and (iv) the mod-12 consecutive-residue transition matrix against the values reported in [14] at comparable scale.

#### 4. THE GLOBAL ANTICORRELATION

TABLE 1. Artin status of consecutive primes, all 50,847,530 pairs with  $7 \leq p_n < p_{n+1} \leq 10^9$ .

| Quantity                                     | Value     |
|----------------------------------------------|-----------|
| $P(\text{Art}_{n+1})$                        | 0.373993  |
| $P(\text{Art}_{n+1} \mid \text{Art}_n)$      | 0.365141  |
| $P(\text{Art}_{n+1} \mid \neg \text{Art}_n)$ | 0.379281  |
| $\delta = \text{difference}$                 | −0.014140 |
| $\phi$ coefficient                           | −0.014140 |
| $\chi^2$ (1 df)                              | 10,167    |
| signed $\sqrt{\chi^2}$ (descriptive)         | −100.8    |

Table 1 presents the central fact: an Artin prime is followed by another Artin prime about 1.4 percentage points less often than a non-Artin prime is. The Pearson statistic is 10,167 (signed square root −100.8 under the nominal independent-status reference model). This is a deterministic finite census, not a random survey; the statistic is reported as a descriptive measure of the strength of association, and should not be interpreted as a sampling confidence level without specifying and justifying a null stochastic model. Consecutive pairs overlap, and residue-cell memberships are serially structured.

The sign is what the Lemke Oliver–Soundararajan repulsion predicts. Heuristically: the Artin property requires  $\left(\frac{10}{p}\right) = -1$ , which is a condition on  $p \pmod{40}$ ; it is further favoured by  $3 \nmid p - 1$ , i.e.  $p \equiv 2 \pmod{3}$ , since a factor of 3 in  $p - 1$  contributes the largest single suppression  $(1 - \frac{1}{3})$  of the primitive-root density among odd primes. Consecutive primes repel in their residues modulo 40 and modulo 3 [14]; conditions favourable to the Artin property therefore tend *not* to repeat at  $p_{n+1}$  when they hold at  $p_n$ , depressing  $P(\text{Art}_{n+1} \mid \text{Art}_n)$ .

For reference, the mod-12 transition matrix in our data displays the repulsion prominently: the diagonal entries  $P(p_{n+1} \equiv r \mid p_n \equiv r)$  are 0.1813–0.1815 for  $r \in \{1, 5, 7, 11\}$ , against the equidistribution value 0.25 — a 27% deficit, consistent in magnitude with the computations in [14] at comparable scales.

## 5. GAP STRUCTURE

Stratifying by the gap  $g = p_{n+1} - p_n$  reveals sign-alternating within-gap effects that are individually far larger in magnitude than the global  $\delta = -0.0141$ . The global value is *not* the pair-frequency-weighted average of the within-gap values: for a marginal conditional risk difference such as  $\delta$ , the strata are additionally reweighted by the gap-conditional distribution of  $\text{Art}_n$ , so no such averaging identity holds. Indeed the pair-weighted mean of the within-gap differences over the 36 qualifying gap classes (50,114,759 pairs, 98.6% of the census) is +0.00159, opposite in sign to the global value. The two quantities answer different questions, and the gap table below should be read as a decomposition of *where* the association lives, not as a summand decomposition of  $\delta$  itself (cf. the caution in Section 6). Table 2 shows all gaps with at least  $10^5$  pairs up to  $g = 60$ ; additional qualifying gaps ( $g \in \{62, 64, 66, 70, 72, 78\}$ ) are included in the analysis scripts and archived results but omitted from the table for brevity. Figure 1 displays  $\delta(g)$  with the residue class of  $g \bmod 40$  marked, including all qualifying gaps.

TABLE 2. Conditional Artin dependence by gap ( $N \geq 10^5$  pairs,  $g \leq 60$ ).  $\delta(g) = P(\text{Art}_{n+1} \mid \text{Art}_n) - P(\text{Art}_{n+1} \mid \neg \text{Art}_n)$ .

| $g$ | $N$       | $P(A A)$     | $P(A \neg A)$ | $\delta(g)$ | $g$ | $N$       | $P(A A)$     | $P(A \neg A)$ | $\delta(g)$ |
|-----|-----------|--------------|---------------|-------------|-----|-----------|--------------|---------------|-------------|
| 2   | 3,424,504 | 0.291        | 0.293         | −0.002      | 32  | 683,896   | 0.176        | 0.386         | −0.209      |
| 4   | 3,424,679 | 0.562        | 0.381         | +0.181      | 34  | 719,531   | 0.445        | 0.445         | +0.001      |
| 6   | 6,089,791 | 0.407        | 0.382         | +0.025      | 36  | 1,171,524 | 0.559        | 0.314         | +0.245      |
| 8   | 2,695,109 | 0.160        | 0.409         | −0.249      | 38  | 548,746   | 0.288        | 0.293         | −0.006      |
| 10  | 3,484,767 | 0.448        | 0.443         | +0.005      | 40  | 648,780   | 0.899        | 0.257         | +0.643      |
| 12  | 4,468,957 | 0.512        | 0.283         | +0.229      | 42  | 954,456   | 0.380        | 0.358         | +0.023      |
| 14  | 2,464,565 | 0.298        | 0.296         | +0.002      | 44  | 389,634   | 0.344        | 0.249         | +0.095      |
| 16  | 1,846,097 | 0.296        | 0.543         | −0.247      | 46  | 334,720   | 0.469        | 0.469         | +0.000      |
| 18  | 3,351,032 | 0.378        | 0.359         | +0.019      | 48  | 577,247   | 0.216        | 0.454         | −0.238      |
| 20  | 1,824,043 | <b>0.000</b> | 0.553         | −0.553      | 50  | 328,066   | 0.311        | 0.307         | +0.005      |
| 22  | 1,569,679 | 0.444        | 0.447         | −0.004      | 52  | 245,804   | 0.572        | 0.381         | +0.191      |
| 24  | 2,367,474 | 0.295        | 0.411         | −0.116      | 54  | 410,754   | 0.382        | 0.356         | +0.026      |
| 26  | 1,119,585 | 0.313        | 0.313         | +0.000      | 56  | 211,462   | 0.225        | 0.389         | −0.164      |
| 28  | 1,219,243 | 0.619        | 0.366         | +0.253      | 58  | 181,948   | 0.437        | 0.441         | −0.004      |
| 30  | 2,177,991 | 0.390        | 0.363         | +0.027      | 60  | 371,839   | <b>0.000</b> | 0.592         | −0.592      |

Three features stand out.

(i) *The exclusion rows.* At  $g = 20$  (1,824,043 pairs) and  $g = 60$  (371,839 pairs),  $P(A \mid A)$  is exactly zero: among 2.2 million consecutive pairs, not one is doubly Artin. This is Theorem 1 in action, and conversely the observation that motivated its proof.

(ii) *The  $g = 40$  spike.*  $P(A \mid A) = 0.899$ , the largest conditional probability in the table and 2.4 times the unconditional density. Two forced mechanisms combine. First, by Corollary 1, if  $p_n$  is Artin then 10 is a non-residue for  $p_{n+1}$  as well, a necessary condition for the Artin property. Second,  $40 \equiv 1 \pmod{3}$  forces the pair’s mod-3 residues:  $p_n \equiv 2 \pmod{3}$  would give  $3 \mid p_n + 40$ , impossible; so every  $g = 40$  pair has  $p_n \equiv 1, p_{n+1} \equiv 2 \pmod{3}$  — that is,  $3 \nmid p_n - 1$  (suppressing  $\text{Art}_n$ ) and  $3 \nmid p_{n+1} - 1$  (favouring  $\text{Art}_{n+1}$ ). Conditioning on  $\text{Art}_n$  therefore selects pairs in which  $p_{n+1}$  enjoys both the non-residue status and the mod-3 advantage. These conditions are consistent

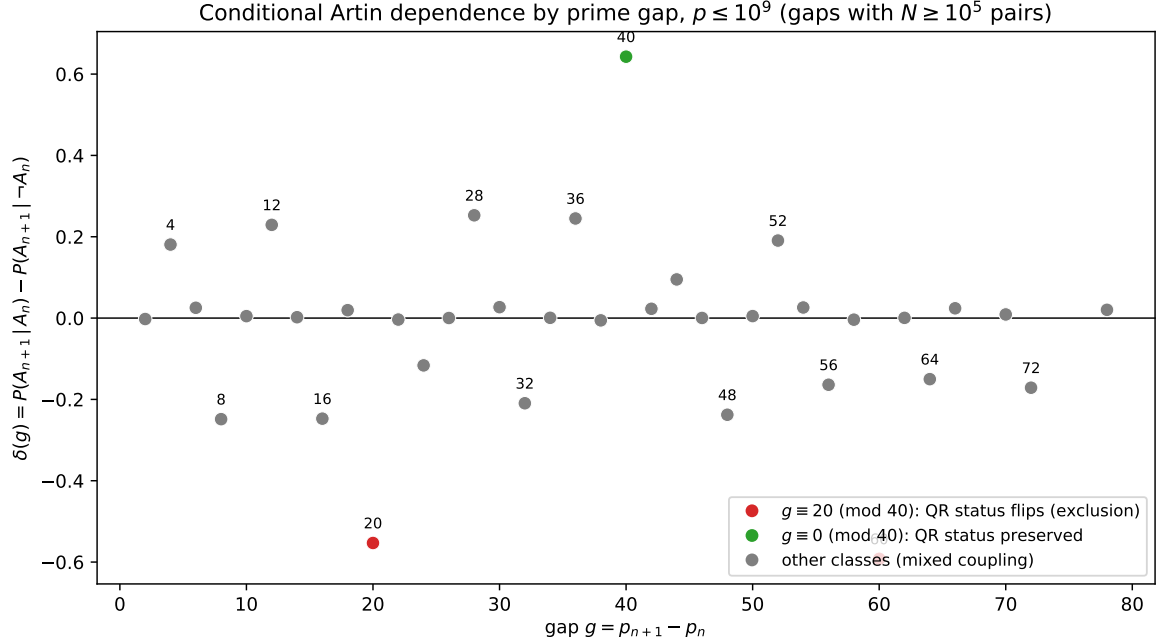


FIGURE 1.  $\delta(g)$  for all gaps with  $N \geq 10^5$  pairs,  $p \leq 10^9$ . Red:  $g \equiv 20 \pmod{40}$  (Theorem 1 forces  $P(A|A) = 0$ ). Green:  $g \equiv 0 \pmod{40}$  (Corollary 1 preserves quadratic-residue status). Grey: other classes.

with the large observed value 0.899 but do not quantitatively derive it without an explicit density calculation.

(iii) *Sign classes.* Several gaps that are divisible by 3 but not constrained mod 40, for instance 6, 18, 30, 42 and 54, show a small positive  $\delta \approx +0.02$ . A natural guess is that the pair shares its mod-3 status, weakly coupling the largest single suppression factor. This is a selected family rather than a rule: other gaps divisible by 3 and lying in neither 0 nor 20 mod 40, such as 12, 24 and 36, do not all behave this way, and we have no statement covering every class. Some gaps with adverse mod-8 and mod-5 shifts show strong negative  $\delta$ . We can therefore offer only a qualitative reading of Table 2 in terms of how the shift by  $g$  permutes the residues of  $p$  modulo 8, 5 and 3; we do not derive the entries, and outside the exclusion classes nothing here is forced.

## 6. RESIDUE CONDITIONING

To what extent does the association persist once the explicit residue channels are held fixed? We treat what follows as an exploratory conditioning diagnostic; it does not answer that question quantitatively. We condition on the *joint* residue class of  $(p_n, p_{n+1})$  to modulus  $m$  and measure the residual within-cell dependence. The statistic reported is the sum of per-cell Pearson  $\chi^2$  values over all non-degenerate cells meeting a minimum count threshold, together with the pair-weighted mean of per-cell  $\delta$ . This is a descriptive aggregate of per-cell associations; it is *not* the standard Cochran–Mantel–Haenszel statistic (which would be a single one-degree-of-freedom test of a common association).

Conditioning on  $m = 120 = \text{lcm}(8, 3, 5)$  fixes the quadratic character  $(\frac{10}{\cdot})$  of both primes (its conductor 40 divides 120) and the mod-3 channel. Cells in which 10 is a quadratic residue for  $p_n$  or  $p_{n+1}$  are degenerate for the analysis (one margin of the  $2 \times 2$  table vanishes — the Artin property is impossible); the informative population is roughly a quarter of all pairs, those in doubly-non-residue cells. The selected populations in Table 3 are the thresholded subsets of this population: 12,255,203

pairs (24.10% of the census) at modulus 120 and 12,086,765 pairs (23.77%) at modulus 840. The unthresholded doubly-non-residue fraction need not be exactly  $1/4$ .

TABLE 3. Selected-cell residual association after conditioning on joint residues mod  $m$ . “ $\delta_{\text{res}}$ ” is the pair-weighted mean within-cell  $\delta$  over non-degenerate cells meeting the count threshold; “cells” is the number of such cells; “pairs used” is the total pairs in those cells. Cells with fewer than 5000 ( $m = 120$ ) or 2000 ( $m = 840$ ) pairs are excluded.

| Conditioning           | cells | pairs used | $\sum \chi^2$ (df) | $\delta_{\text{res}}$ |
|------------------------|-------|------------|--------------------|-----------------------|
| none (global)          | 1     | 50,847,530 | 10,167 (1)         | −0.014140             |
| mod 120 joint residues | 145   | 12,255,203 | 789.2 (145)        | −0.001055             |
| mod 840 joint residues | 633   | 12,086,765 | 719.8 (633)        | −0.000473             |

**Interpretation and limitations.** The three statistics in Table 3 have successively smaller absolute values:  $-0.0141$  (global) then  $-0.00106$  (mod 120, 145 cells covering 12.3 million pairs) then  $-0.00047$  (mod 840, 633 cells covering 12.1 million pairs). We emphasise that these are three different quantities computed on three different populations, not one quantity being reduced; the comparison is an exploratory conditioning diagnostic, not a decomposition. Note also that a signed mean can be small because sizable positive and negative within-cell associations cancel, so smallness in this metric is not by itself evidence that dependence has gone. The summed per-cell  $\chi^2$  at mod 840 is 719.8 on 633 nominal degrees of freedom ( $\chi^2/\text{df} = 1.14$ ); we do not read  $\chi^2/\text{df} \approx 1$  as the disappearance of dependence, since no reference distribution has been established for this serially structured, selected collection. Two further observations: (i) the three values decrease in magnitude in the order stated ( $-0.0141$ ,  $-0.00106$ ,  $-0.00047$ ); (ii) a split-sample check (pairs with  $p < 5 \times 10^8$  versus  $p \geq 5 \times 10^8$ , conditioning mod 120) gives residuals  $-0.00096$  and  $-0.00094$  — stable across the range.

Several caveats apply. The global  $\delta$  over 50.8 million pairs and the selected-cell  $\delta_{\text{res}}$  over 12.1 million selected informative pairs are *different estimands on different populations*; arithmetic comparison of their magnitudes is descriptive but does not constitute a proper decomposition. A marginal risk difference is not, in general, the pair-weighted mean of stratum risk differences without a between-group term and specified weights. The per-cell Pearson calibration assumes independent observations within each cell, which is not established for this serially structured deterministic census. The paper does not specify a numerical model for the omitted channels against which “compatible” could be formally tested. Those omitted channels are not only the divisibility channels at  $q \geq 11$ : fixing  $p$  modulo 840 determines whether 3, 5 and 7 divide  $p - 1$ , but not whether  $10^{(p-1)/q} \equiv 1 \pmod{p}$  when one of them does, so cubic and higher power-residue obstructions at these same small primes are also unmodelled. The mod-3 channel here is the divisibility channel, not the full cubic-residue condition.

Joint-residue conditioning yields a smaller selected-cell signed residual association than the global figure. The remaining association is small in that metric but is not established to vanish. These results are consistent with a substantial role for arithmetic residue structure; they do not constitute a complete decomposition or an independent quantitative verification of the Hardy–Littlewood-based explanation of the consecutive-prime bias.

## 7. THE INTERMEDIATE LINK: REPULSION OF $\omega(p - 1)$

The chain from residue bias to Artin status passes through the factorisation of  $p - 1$ . Its first summary statistic is  $\omega(p - 1)$ , the number of distinct prime factors, which has normal order  $\log \log p$  over shifted primes  $p - 1$  — a classical result of Erdős [4]. (We invoke the normal order only; we do

not use a mean-value asymptotic, which would need separate tail control.) If consecutive primes repel modulo every small  $q$ , then  $q \mid p_n - 1$  makes  $q \mid p_{n+1} - 1$  *less* likely than average, and summing over  $q$  one predicts a negative correlation between  $\omega(p_n - 1)$  and  $\omega(p_{n+1} - 1)$ .

The data confirm this with a value that is, to our knowledge, unreported:

$$r(\omega(p_n - 1), \omega(p_{n+1} - 1)) = -0.0410 \quad (50,847,530 \text{ pairs, } p \leq 10^9).$$

The factorisations of  $p - 1$  for consecutive primes are therefore not independent. We stress that this measurement establishes dependence, not its cause. The heuristic above accounts only for the same- $q$  terms: writing  $I_q(p) = \mathbf{1}_{q \mid p-1}$ ,

$$\text{Cov}(\omega(p_n - 1), \omega(p_{n+1} - 1)) = \sum_q \text{Cov}(I_q(p_n), I_q(p_{n+1})) + \sum_{q \neq r} \text{Cov}(I_q(p_n), I_r(p_{n+1})),$$

and the cross terms  $q \neq r$  are neither estimated nor signed here. Summing fixed-modulus predictions over all  $q$  up to a growing cutoff would in any case require uniformity and tail control that we do not supply. Nor is  $\omega$  a sufficient statistic for Artin status: which primes divide  $p - 1$ , and whether 10 is a  $q$ th power residue, both matter. The observed repulsion is consistent with the same residue-class repulsion that governs the primes, but we do not establish that it has the same cause. The Pearson coefficient is reported as a descriptive nominal value.

## 8. OPEN QUESTIONS

The Hardy–Littlewood-based framework of [14] makes the residue transition frequencies of consecutive primes quantitatively predictable, and every channel in our conditioning is built from such frequencies. This motivates the following open questions.

**Conjecture 1** (Persistence). *For primes up to  $x$ , the consecutive-pair Artin correlation  $\delta(x) = P(\text{Art}_{n+1} \mid \text{Art}_n) - P(\text{Art}_{n+1} \mid \neg \text{Art}_n)$  is negative for all sufficiently large  $x$ .*

We have measured  $\delta$  at four decade-spaced cutoffs:

$$\delta(10^8) = -0.013360, \quad \delta(10^9) = -0.014140, \quad \delta(10^{10}) = -0.014723, \quad \delta(10^{11}) = -0.014998,$$

the last two from an independent recomputation described in Section 3 (over 455,052,508 and 4,118,054,810 primes respectively). At these four cutoffs the magnitude of  $\delta$  increases, and the increments  $-0.000780$ ,  $-0.000583$ ,  $-0.000275$  decrease. No turnover is observed over the measured range.

We draw no asymptotic conclusion from this. Four nested cumulative cutoffs are not four independent observations, and they determine neither an eventual sign nor a limit nor a decay rate. Two specific negative remarks are all that the values support: the ratio of successive increments is not constant (0.75 then 0.47), so a geometric increment model does not describe them — extrapolating geometrically from the first three values predicts  $\delta(10^{11}) \approx -0.01516$  against a measured  $-0.014998$ , a discrepancy of  $1.6 \times 10^{-4}$  — and  $\delta(x) \log x$  is not constant ( $-0.246$ ,  $-0.293$ ,  $-0.339$ ,  $-0.380$ ), so a bare  $c/\log x$  law does not describe them either.

It is worth recording explicitly that monotone strengthening with decelerating increments does *not* constrain the eventual rate, and in particular does not exclude decay to zero. Writing  $t = \log_{10} x$ , the function

$$f(t) = -\frac{0.248463}{t} - \frac{1.495239}{t^2} + \frac{41.285554}{t^3} - \frac{162.098640}{t^4}$$

reproduces all four measured values at  $t = 8, 9, 10, 11$ , is negative for all  $t \geq 8$ , strengthens across the measured range, and yet satisfies  $f(t) \sim -0.248463/t$ , so that  $f \rightarrow 0$ ; its turning point lies at  $t \approx 11.56$ , just beyond the last cutoff. This is an interpolation constructed to make the point, not a proposed model, and we attach no significance to its coefficients. It shows only that our data are

equally compatible with eventual decay to zero and with persistence of a nonzero correlation, and that the question is not settled by extending the census a decade or two further.

We also note that the conjecture as stated — that  $\delta$  is eventually negative — is compatible with  $\delta(x) \rightarrow 0$ ; eventual negativity and decay to zero are not competing alternatives. A separate reason for caution against assuming a nonzero limit arises in a restricted setting. Consider a *separable* fixed-modulus channel model: write  $R, S$  for the residue classes of  $(p_n, p_{n+1})$  modulo a fixed  $M$ , and suppose the modelled indicators satisfy  $\mathbb{E}[X \mid R, S] = \alpha(R)$ ,  $\mathbb{E}[Y \mid R, S] = \beta(S)$  and  $\text{Cov}(X, Y \mid R, S) = 0$ , so that the only modelled dependence between the two endpoints runs through the joint law of  $(R, S)$ . If that joint law tends to the product of its marginals, as the leading term of the Lemke Oliver–Soundararajan prediction gives for fixed  $M$  [14], then  $\mathbb{E}[XY] - \mathbb{E}[X]\mathbb{E}[Y] \rightarrow 0$ , and hence the modelled  $\delta \rightarrow 0$  whenever the limiting source marginal is nondegenerate. Non-uniform singular-series weights across gap classes do not by themselves defeat this conclusion: the gaps grow with  $x$ , and it is the surviving total covariance, rather than the non-uniformity of the weights, that would have to be estimated.

The separability hypothesis is essential and is not automatic. Conditioning on the joint pair  $(R, S)$  alone does not force independence: taking  $R, S$  independent and uniform on the sixteen reduced classes modulo 40 and setting  $X = Y = Q(R)Q(S)$ , where  $Q$  indicates the eight classes on which the relevant quadratic symbol is  $-1$ , gives perfectly equidistributed residue pairs together with  $\mathbb{E}[X] = \mathbb{E}[Y] = \mathbb{E}[XY] = \frac{1}{4}$  and  $\text{Cov}(X, Y) = \frac{3}{16}$ . So the observation above constrains separable finite-modulus models, not every model built on a fixed modulus. Extending a finite-obstruction approach to a statement about the limit would in addition require joint-distribution information at each finite channel together with a uniform bound on the omitted higher-power-residue obstructions; the quadratic channel visible here is only the first of these. The question of the limiting behaviour is unresolved; finite-range measurements cannot distinguish the two alternatives.

**Research question (residue completeness).** Does there exist, for every  $\varepsilon > 0$ , a modulus  $M$  such that conditioning on the joint residues of  $(p_n, p_{n+1})$  modulo  $M$  reduces some appropriate within-cell dependence measure below  $\varepsilon$  in absolute value? This is stated as a question rather than a conjecture because a precise formulation requires specifying the order of limits (over  $x$  and  $M$ ), the treatment of sparse and degenerate cells, the dependence criterion (e.g. a covariance decomposition rather than a signed cell mean), and the range of  $x$  over which the bound is asserted. We do not know a formulation that avoids these ambiguities.

We remark that a sufficiently uniform Hardy–Littlewood prime  $k$ -tuple framework, together with a justified treatment of the no-intermediate-prime restriction, would be the natural setting in which to predict the joint distribution of  $(p_n \bmod M, p_{n+1} \bmod M, g)$  via singular series. The pair conjecture alone does not suffice, and the restriction is not an optional refinement: consecutiveness is part of the event, and its indicator

$$\mathbf{1}_{p \text{ prime}} \mathbf{1}_{p+g \text{ prime}} \prod_{1 \leq h < g} (1 - \mathbf{1}_{p+h \text{ prime}})$$

expands into higher-tuple correlations that two-point counts do not determine. This is why [14] proceeds from a  $k$ -tuple heuristic rather than from the pair conjecture. Whether such predictions carry the Chebotarev and power-residue information needed for Artin status is a further question again, and requires analysis we do not attempt here.

**Conjecture 2** ( $\omega$  repulsion).  *$r(\omega(p_n - 1), \omega(p_{n+1} - 1))$  is negative for all large  $x$  and tends to 0 as  $x \rightarrow \infty$ .*

The single reported value does not establish eventual sign or limit. We state this as a conjecture consistent with the data.

## 9. DISCUSSION

**What is new here.** The individual ingredients are classical: quadratic reciprocity, the quadratic-residue obstruction to primitive roots, and the consecutive-prime bias of [14]. The contributions of this paper are (i) the observation that these combine into a deterministic exclusion law for prime pairs at gaps  $\equiv 20 \pmod{40}$  (Theorem 1) — elementary, but we have not found it stated elsewhere; (ii) a measurement of the consecutive-prime Artin correlation, its gap profile, and the  $\omega$  repulsion; and (iii) evidence that arithmetic residue structure is substantially connected to the measured dependence. The theorem is an immediate application of standard quadratic reciprocity; its simplicity is a feature, not a limitation.

**Relation to runs of Artin primes.** Pollack [17] proved (under GRH) that for every nonsquare  $g \neq -1$  and every  $m$ , there are infinitely many runs of  $m$  consecutive primes each having  $g$  as a primitive root. Baker and Pollack [2] proved unconditionally that if  $Q$  contains at least  $\exp(Cm)$  multiplicatively independent primes, then infinitely many runs of  $m$  consecutive primes each have some element of  $Q$  as a primitive root, with the primes lying in a bounded-length interval. Our results add an arithmetic constraint to the base-specific version: for base 10, no run of all-Artin consecutive primes can cross a gap  $\equiv 20 \pmod{40}$ . We have not inferred longer-run frequencies from the pair statistic.

**Future work.** Three extensions suggest themselves. (1) Other bases: the same pipeline with  $a = 2, 3, 5, 6, 7$  would test the conductor prediction of the Remark after Corollary 1 (e.g. conductor 8 for  $a = 2$ , so exclusion classes mod 8; conductor 24 for  $a = 6$ ). (2) Scale: extending to  $10^{12}$  would help distinguish decay from persistence. (3) Theory: a derivation of  $\delta(g)$  from the singular series of [9] combined with the density computations of Hooley’s method, at least for the dominant channels, appears feasible and would replace our empirical conditioning with a formula.

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## DATA AVAILABILITY

All code required to reproduce every number in this paper is supplied in the accompanying reproduction package (C segmented sieve, Python analysis scripts, figure script, and raw output logs). The code, manuscript, results and Lean development are at `repositoryURLwithheldforreview`, a repository dedicated to this paper; the full per-prime CSV is not included there. That package

contains both implementations of the census described above, so a reader can repeat the independent check as well as the original computation. The sieve limit is configurable via a command-line argument, and the scripts accept their input and output paths as arguments, defaulting to repository-relative locations. The per-prime dataset itself (1.8 GB CSV covering all 50,847,531 primes from 7 to 999,999,937: gap,  $\omega(p-1)$ , Artin indicator, and small residues for each prime) is not deposited because of its size, but it is exactly and deterministically reproducible from the included C source on a single machine; it is also available from the author on request.

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